

Sequential Tradeability Testing for Alpha Signals

A Bayesian optimal stopping view of alpha validation

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When should a quant team stop researching a signal and admit it to live trading?

The object of the project

research evidence arrives over time $\implies$ continue, activate, or reject

Thesis

Alpha validation is not only a fixed-horizon hypothesis test. It is a sequential real-options problem: more evidence has value, but waiting is costly.

The project uses the EKV drift-learning model as the filtering state, then changes the terminal reward from statistical accuracy to economic tradeability.

Why the usual test is not the right decision problem

Fixed-horizon test

at T : is the signal significant?

- ▶ ignores the option value of stopping early;
- ▶ ignores the cost of waiting;
- ▶ treats weak, dead, and expensive signals too similarly.

Sequential admission

τ : is it worth trading yet?

- ▶ stop when enough is known;
- ▶ reject when the posterior dead state is too likely;
- ▶ activate only after uncertainty and costs.

Storyline

1. EKV: learn an unknown drift from noisy Brownian evidence.
2. Key identity: posterior uncertainty is also the speed of learning.
3. Project move: replace EKV's terminal statistical loss with an economic alpha payoff.
4. Stress test: Gaussian beliefs work for live signals, but over-activate dead ones.
5. Extension: a three-state dead/alive prior makes false discovery explicit.
6. Evidence: OSAP and WRDS experiments test whether the rule is useful in data.

EKV starts with drift learning

$$Y_t = X t + W_t, \quad t \geq 0.$$

- ▶ X is an unknown constant drift drawn from a prior μ .
- ▶ Y_t is cumulative noisy evidence.
- ▶ $\mathcal{F}_t^Y = \sigma(Y_s : 0 \leq s \leq t)$ is the information available to the decision maker.

Alpha interpretation

X becomes unknown alpha strength, and Y_t becomes cumulative normalized signal-aligned evidence.

Bayes' rule gives the posterior state

$$\mu_{t,y}(du) = \frac{\exp\{uy - u^2t/2\}\mu(du)}{\int_{\mathbb{R}} \exp\{vy - v^2t/2\}\mu(dv)}.$$

$$\widehat{X}_t = G(t, Y_t)$$

$$G(t, y) = \mathbb{E}[X \mid Y_t = y]$$

$$H(t, y) = \text{Var}(X \mid Y_t = y)$$

$$\partial_y G(t, y) = H(t, y)$$

The posterior mean and posterior variance are the low-dimensional state variables.

The filtering identity

$$\widehat{W}_t = Y_t - \int_0^t \widehat{X}_s ds$$

$$d\widehat{X}_t = \Psi(t, \widehat{X}_t) d\widehat{W}_t, \quad \Psi(t, \widehat{X}_t) = \text{Var}(X \mid \mathcal{F}_t^Y).$$

Intuition

The same quantity, posterior uncertainty, is both the remaining estimation error and the local volatility of the posterior mean. High uncertainty means there is still a lot to learn.

EKV's optimal stopping problem

$$\inf_{\tau} \mathbb{E} \left[(X - \widehat{X}_{\tau})^2 + c\tau \right].$$

EKV ask: when should one stop observing and estimate the drift?

$$\mathbb{E}[\Psi(\tau, \widehat{X}_{\tau})] = \text{Var}(X) - \mathbb{E} \left[\int_0^{\tau} \Psi^2(s, \widehat{X}_s) ds \right].$$

The estimation loss decreases exactly by accumulated learning speed.

The EKV problem becomes a running-cost problem

$$\inf_{\tau} \mathbb{E} \left[\int_0^{\tau} (c - \Psi^2(s, \widehat{X}_s)) \, ds \right].$$

Cost of waiting

c

observation, time, capacity

Benefit of waiting

Ψ^2

expected reduction in
posterior variance

Two EKV benchmarks we use to validate the math

Gaussian prior

$$\begin{aligned} X &\sim N(m_0, q_0) \\ q(t) &= \frac{q_0}{1 + q_0 t}, \quad dm_t = q(t) d\widehat{W}_t. \\ \tau^* &= \left(\frac{1}{\sqrt{c}} - \frac{1}{q_0} \right)^+. \end{aligned}$$

Symmetric Bernoulli prior

$$\begin{aligned} X &\in \{-\beta, \beta\} \\ \Psi(x) &= \beta^2 - x^2. \end{aligned}$$

Stop when the posterior mean exits a symmetric interval.

The project changes the terminal reward

EKV: estimate accurately $\implies$ ours: trade only if useful

Important distinction

The EKV theorem does not directly prove our finance boundary. EKV supplies the posterior dynamics. We define a new economic reward and solve the corresponding optimal stopping problem.

Variable translation

EKV object	Alpha validation object
X	unknown alpha strength
Y_t	cumulative normalized evidence
$\widehat{X}_t$	posterior mean alpha
Ψ	posterior uncertainty and learning speed
c	research cost, delay, opportunity cost
τ	activation or rejection time

Economic activation payoff

$$mz - \frac{\lambda}{2}(\sigma_r^2 + q)z^2 - \kappa.$$

At stopping, m is posterior mean alpha and q is posterior parameter risk. Optimizing over exposure z gives

$$\Phi(m, q) = \left(\frac{m^2}{2\lambda(\sigma_r^2 + q)} - \kappa \right)^+.$$

$$|m| > \sqrt{2\lambda(\sigma_r^2 + q)\kappa}$$

is the immediate activation threshold, but not the dynamic stopping boundary.

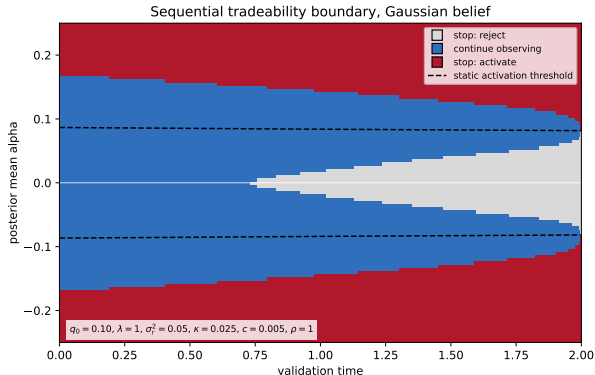
The finite-horizon stopping problem

$$V(t, m) = \sup_{\tau \in [t, T]} \mathbb{E}_{t, m} \left[e^{-\rho \tau} \Phi(m_\tau, q(\tau)) - \int_t^\tau c \, ds \right].$$

Here ρ captures alpha decay or opportunity loss during validation.

$$\max \left\{ e^{-\rho t} \Phi(m, q(t)) - V, V_t + \frac{1}{2} q(t)^2 V_{mm} - c \right\} = 0.$$

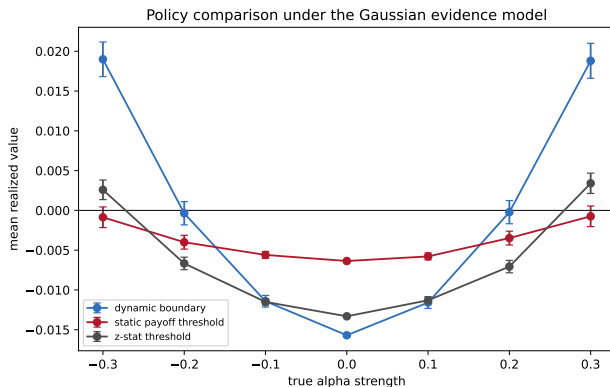
What the boundary means



- ▶ Red: stop and activate.
- ▶ Gray: stop and reject.
- ▶ Blue: continue observing.
- ▶ Dashed curves: static activation threshold.

The continuation region is not the set where payoff is positive. It is where one more observation has positive option value.

First stress test: Gaussian beliefs



What works

For large $|\theta|$, the dynamic boundary captures learning option value and improves realized economic value.

What fails

At $\theta = 0$, the Gaussian prior activates about 90.1% of null paths.

Why this failure is informative

$$X \sim N(0, q_0) \quad \Rightarrow \quad \mathbb{P}(X = 0) = 0.$$

In finance, many researched signals are exactly dead in the economic sense: they are not small live alphas, they are false discoveries or implementation artifacts.

Modeling implication

A signal-admission model should assign posterior mass to the event “this signal has no alpha.”

A three-state prior for signal research

$$\theta \in \{-a, 0, +a\}.$$

State	Interpretation
$-a$	live short signal
0	dead signal
$+a$	live long signal

This prior is not a cosmetic change. It changes what the posterior can learn.

Posterior probabilities under the dead/alive prior

$$\pi_i(t, y) = \frac{p_i \exp\{\theta_i y - \theta_i^2 t/2\}}{\sum_j p_j \exp\{\theta_j y - \theta_j^2 t/2\}}.$$

$$m(t, y) = \sum_i \pi_i(t, y) \theta_i, \quad q(t, y) = \sum_i \pi_i(t, y) \theta_i^2 - m(t, y)^2.$$

If y stays near zero while time passes, $\pi_0(t, y)$ rises because live alphas would likely have generated directional evidence.

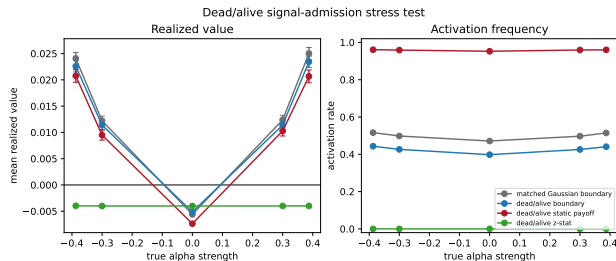
Dead-state penalty

$$\Phi_{\eta}(t, y) = e^{-\rho t} \left(\frac{m(t, y)^2}{2\lambda(\sigma_r^2 + q(t, y))} - \kappa - \eta\pi_0(t, y) \right)^+.$$

The penalty $\eta\pi_0(t, y)$ prices the cost of admitting a false discovery.

$$\max \left\{ \Phi_{\eta}(t, y) - V, V_t + m(t, y)V_y + \frac{1}{2}V_{yy} - c \right\} = 0.$$

Controlled validation of the dead/alive rule



Null activation

47.1% $\rightarrow$ 39.8%

matched Gaussian boundary versus
dead/alive boundary.

Null realized value

-0.00560 $\rightarrow$ -0.00513

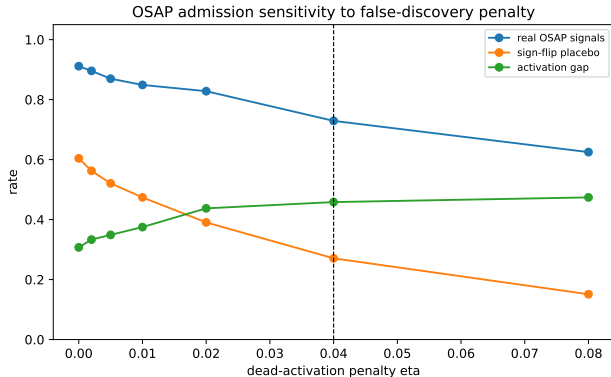
The cost is slightly lower live-alpha value, which is the expected false-discovery versus power tradeoff.

Real-data design

Data	Chen-Zimmermann OSAP long-short monthly returns
Eligible signals	192 predictors with enough history
Calibration	120 months
Sequential validation	up to 240 months
Holdout	120 post-decision months
Evidence increment	$\Delta Y_t = (r_t / \hat{\sigma}_m) \sqrt{dt}$
Placebo	sign-flipped validation returns

Under a null with monthly variance $\hat{\sigma}_m^2$, $\text{Var}(\Delta Y_t) \approx dt$, so evidence is on Brownian scale.

Penalty calibration with real signals and placebos



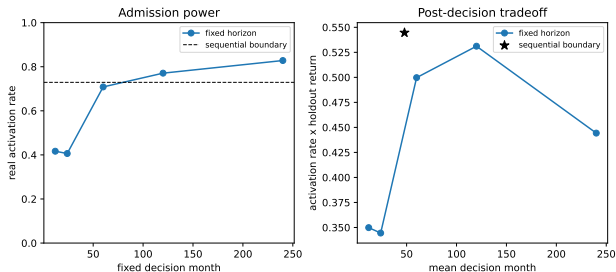
At $\eta = 0.04$

real activation = 72.9%

placebo activation = 27.1%

Higher false-discovery penalties reduce both rates, but reduce placebo activation faster.

Is the rule really sequential?



Sequential rule

72.9% real activation

47.9 month average decision time

0.545 activation-weighted holdout

Fixed 120- and 240-month tests activate more signals, but require much longer waits and have weaker activation-weighted holdout returns.

WRDS stock-level reconstruction

Return stream	N	Active	Standard active	Std. holdout
Gross stock-level	3	3	3	1.227%
Net range-cost stress	3	3	0	—

The gross result says the reconstructed signals are tradeable before costs. The net range-cost result says the conclusion is fragile under a harsh cost stress test.

This is the right empirical tension for the project: admission must survive uncertainty and implementation frictions.

Mathematical and numerical checks

- ▶ Posterior probabilities remain positive and sum to one.
- ▶ $\partial_y G(t, y) = H(t, y)$ and $\partial_y m(t, y) = q(t, y)$ are checked numerically.
- ▶ Gaussian posterior precision update matches the closed form.
- ▶ The obstacle solver satisfies terminal value, symmetry, and monotonicity tests.
- ▶ Dead/alive finite-difference diagnostics: zero obstacle violation, zero terminal error, symmetry error 4.2×10^{-16} .
- ▶ Grid convergence: the paper grid has $V(0, 0)$ error below 9×10^{-5} relative to the fine grid.

What is novel here?

- ▶ The EKV filtering state is used for alpha validation rather than drift estimation alone.
- ▶ The terminal payoff is economic: posterior alpha must clear risk, uncertainty, and costs.
- ▶ The dead/alive prior makes false discovery a state variable, not just an after-the-fact error.
- ▶ The empirical test asks whether sequential stopping beats fixed-horizon posterior tests at matched placebo activation.

One-line conclusion

Alpha validation is a sequential admission problem, not just a significance test.

Final takeaways

1. EKV gives a rigorous Bayesian learning state.
2. Economic tradeability changes the stopping reward.
3. Gaussian beliefs reveal the learning option value, but miss dead alphas.
4. A dead/alive prior reduces false activation in simulation and placebo tests.
5. Real data suggest the sequential rule admits useful signals earlier than fixed-horizon tests.

References



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